Matrix and FFT solutions

A13-M01 Dimensions and multiplication

$AB = [4, 5; 10, 11]$ and $BA = [1, 2, 3; 4, 5, 6; 5, 7, 9]$. Entry $c_{21} = 4(1) + 5(0) + 6(1) = 10$. There are $2 \cdot 2 \cdot 3 = 12$ scalar multiplications.

A13-M02 Block multiplication

$A_1 : 4 \times 2, A_2 : 4 \times 4, B_1 : 2 \times 5, B_2 : 4 \times 5$. Both products are 4×5. Split each entry sum into indices 1–2 and 3–6 to obtain $AB = A_1B_1 + A_2B_2$.

A13-M03 A zero-pivot counterexample

The first divisor is zero. Set $P = [0, 1; 1, 0], L = I, U = [1, 1; 0, 1]$. Then $PA = LU$. One exchange gives $\det A = -1(1)(1) = -1 \neq 0$, so the matrix is invertible.

A13-M04 Trace two pivot steps

Swap rows 1 and 3 to get $[5, 6, 3; 3, 4, 4; 1, 2, 0]$. Eliminate to get $[5, 6, 3; 0, 2/5, 11/5; 0, 4/5, -3/5]$. Swap rows 2 and 3 to get $[5, 6, 3; 0, 4/5, -3/5; 0, 2/5, 11/5]$. Eliminate with multiplier 1/2 to get $U = [5, 6, 3; 0, 4/5, -3/5; 0, 0, 5/2]$. The first column of L must become $(1, 1/5, 3/5)^T$ or $PA = LU$ fails.

Round 2: swap the working matrix, P and stored L

U_1

$$\begin{bmatrix} 5 & 6 & 3 \\ 0 & 2/5 & 11/5 \\ 0 & 4/5 & -3/5 \end{bmatrix}$$

P_1

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

L: stored column 1

$$\begin{bmatrix} 1 & 0 & 0 \\ 3/5 & 1 & 0 \\ 1/5 & 0 & 1 \end{bmatrix}$$

$|4/5| > |2/5| \rightarrow R_2 \leftrightarrow R_3$

Before the swap; exchange only the completed column of L.

A13-M05 Triangular solves and reuse

$Pb = (8, 3, 7)^T$. Forward substitution gives $y = (8, 7/5, 3/2)^T$. Back substitution gives $x_3 = 3/5, x_2 = 11/5, x_1 = -7/5$, with $Ax = b$. One factorization and q pairs of solves cost $\Theta(n^3 + qn^2)$.

A13-M06 Determinant and one inverse column

Two exchanges give $\det A = 10$. For e_1, $Pe_1 = (0, 1, 0)^T$ and $y = (0, 1, -1/2)^T$. Back substitution gives $z_3 = -1/5, z_2 = 11/10, z_1 = -6/5$. Check $Az = e_1$. A full inverse requires n solves, costing $\Theta(n^3)$ after factorization, compared with $\Theta(n^2)$ for one right-hand side.

A13-M07 Verify the complete least-squares fit

$A = [1, 0; 1, 1; 1, 2]$, $A^TA = [3, 3; 3, 5]$, and $A^Tb = (5, 6)^T$. The solution is $(7/6, 1/2)^T$, predictions are $(7/6, 5/3, 13/6)^T$, residuals are $(-1/6, 1/3, -1/6)^T$, and squared error is 1/6. Both $\sum r_i = 0$ and $\sum t_i r_i = 0$. Scatterplot residuals are vertical at fixed horizontal coordinates, not shortest point-line distances. Orthogonality to $\mathrm{col}(A)$ holds in three-dimensional data space.

A13-M08 Conditioning and the limits of QR

If the singular values are $\sigma_i > 0$, then A^TA has eigenvalues σ_i^2, so the extreme ratio is $(\sigma_{\max}/\sigma_{\min})^2$. The example gives 10^4 and 10^8. Householder QR uses orthogonal transformations and a triangular solve without forming A^TA. It avoids that artificial squaring but retains sensitivity caused by nearly rank-deficient data.

A13-X01: A block inverse

$B = 2, C = 1, D = 3, E = 4$ and $S = 4 - 3(1/2)(1) = 5/2$. The formula gives $A^{-1} = \frac{1}{5}\begin{bmatrix} 4 & -1 \\ -3 & 2 \end{bmatrix}$; multiplying yields $\frac{1}{5}[5, 0; 0, 5] = I$. No: $[0, 1; 1, 0]$ is invertible despite a zero top-left block, so change the partition or exchange rows.

A13-X02: Positive definiteness and Cholesky

$l_{11} = 2, l_{21} = 2/2 = 1, l_{22} = \sqrt{5-1} = 2$; hence $L = [2, 0; 1, 2]$ and $LL^T = A$. Since L is invertible, $x^T Ax = \|L^T x\|^2 > 0$ for nonzero x. Solve $Ly = (6, 7)^T$ to obtain $y = (3, 2)^T$; then $L^T x = y$ gives $x = (1, 1)^T$. Substitution gives $Ax = (6, 7)^T$.

A13-X03: Multiplication from inversion

$T = [1, 2, 0; 0, 1, 3; 0, 0, 1]$ and $T^{-1} = [1, -2, 6; 0, 1, -3; 0, 0, 1]$. Their product is the identity, and the upper-right entry is $XY = 6$. Generally $I(n) = \Omega(n^2)$ absorbs construction cost and $I(3n) = O(I(n))$ absorbs the dimension increase, proving $M(n) = O(I(n))$.

A13-X04: Cost of inversion

The principal block and Schur complement of an SPD matrix are SPD, so both recursive inverses exist. At level i, work is $O(2^i(n/2^i)^\omega) = O(n^\omega 2^{-(\omega-1)i})$. Since $\omega > 2$, the series converges, giving $O(n^\omega)$ including leaf costs. For real invertible A, $A^T A$ is SPD and $A^{-1} = (A^T A)^{-1} A^T$. Forming $A^T A$ squares the 2-norm condition number and may amplify floating-point error.

A13-X05: Expand Strassen's formula

$P_3 + P_5 = A_{11}B_{12} - A_{11}B_{22} + A_{11}B_{22} + A_{12}B_{22} = A_{11}B_{12} + A_{12}B_{22} = C_{12}$. Here $P_3 = 1(6 - 8) = -2$ and $P_5 = (1 + 2)8 = 24$, so $C_{12} = 22$. The derivation uses distributivity without changing multiplication order, so it also holds for compatible matrix blocks.

A13-X06: Rank-deficient least squares

$\text{col}(A)$ is spanned by $u = (1, 2, 0)^T$, so the projection is $p = u(u^T b)/(u^T u) = (8/5, 16/5, 0)^T$. The residual $r = b - p = (2/5, -1/5, 1)^T$ satisfies $A^T r = (0, 0)^T$. Since $Ax = p$ is equivalent to $x_1 + x_2 = 8/5$, all solutions are $(4/5, 4/5)^T + t(1, -1)^T$. Their squared norms are $32/25 + 2t^2$, so the unique minimum-norm solution is $(4/5, 4/5)^T$.

A13-X07: Elementwise Cholesky

$l_{11} = 3, l_{21} = 1, l_{31} = 2; l_{22} = \sqrt{5-1} = 2, l_{32} = (4 - 2 \cdot 1)/2 = 1; l_{33} = \sqrt{9 - 2^2 - 1^2} = 2$. Thus $L = [3, 0, 0; 1, 2, 0; 2, 1, 2]$, and direct multiplication gives $LL^T = A$. As L has nonzero diagonal, every nonzero x satisfies $x^T Ax = \|L^T x\|^2 > 0$; this A is SPD.

A13-X08: Sufficiency of the normal equation

$A\hat{x} = (8/5, 16/5, 0)^T$ and $r = b - A\hat{x} = (2/5, -1/5, 1)^T$, so $A^T r = (0, 0)^T$. For $z = (u, v)^T$, $Az = (u + v, 2(u + v), 0)^T$. The cross term vanishes, giving $f(\hat{x} + z) - f(\hat{x}) = \|Az\|^2 = 5(u + v)^2 \geq 0$. Equality holds exactly when $u + v = 0$; all minimizers are $\hat{x} + t(1, -1)^T$.

A13-X09: A second pivot and triangular solves

The first column selects 2 without a row swap. Multipliers $l_{21} = 1/2, l_{31} = 0$ produce $W = [2, 0, 1; 1/2, 0, 3/2; 0, 3, 1]$ and $p = (1, 2, 3)$. The second column selects 3; swapping whole rows 2 and 3 gives $W = [2, 0, 1; 0, 3, 1; 1/2, 0, 3/2]$ and $p = (1, 3, 2)$. The second multiplier $l_{32} = 0$ leaves the trailing entry unchanged. Thus $L = [1, 0, 0; 0, 1, 0; 1/2, 0, 1]$, $U = [2, 0, 1; 0, 3, 1; 0, 0, 3/2]$, and $LU = PA = [2, 0, 1; 0, 3, 1; 1, 0, 2]$.

The row swap must exchange old $l_{21} = 1/2, l_{31} = 0$ as well; otherwise these multipliers refer to the wrong rows and $LU \neq PA$. Here $Pb = (4, 5, 5)^T$; forward substitution gives $y = (4, 5, 3)^T$ and back substitution gives $x_3 = 2$, $x_2 = 1$, $x_1 = 1$. Check $Ax = (4, 5, 5)^T$.

A13-X10: Singularity and cost

The first column selects 2 and swaps rows, giving $[2, 4; 1, 2]$; multiplier $1/2$ updates the trailing entry to $2 - (1/2)4 = 0$. The second column has no nonzero candidate, so exact arithmetic reports singularity; indeed $\det A = 0$. A positive tolerance only indicates failure under that rule: $\mathrm{diag}(1, \epsilon)$ is invertible for nonzero ϵ even if ϵ falls below the threshold.

Counting each addition, subtraction, multiplication or division as one flop, LUP costs $2n^3/3 + O(n^2)$ and Cholesky $n^3/3 + O(n^2)$; the n square roots may be priced separately. Partial pivoting additionally uses $O(n^2)$ comparisons and at most n row swaps moving $O(n^2)$ entries, separate from multiply-add flops.

A13-F01

Every output is 1, since only x_0 is nonzero.

A13-F02

Three-bit $3 = 011$ reverses to $110 = 6$; order is $[0, 4, 2, 6, 1, 5, 3, 7]$.

A13-F03

Reorder to $[1, 0, 2, -1]$. Length 2 with twiddle 1 gives $[1, 1, 1, 3]$. Length 4: $(0, 2)$ uses 1 and gives $(2, 0)$; $(1, 3)$ uses i and gives $(1 + 3i, 1 - 3i)$.

A13-F04

The inner sum $\sum_{k=0}^{n-1} \omega_n^{k(r-j)}$ is n for $r = j$ and otherwise zero by the geometric-series formula. Multiplication by $1/n$ leaves x_j.

A13-F05

Reorder to $[2, 0, 1 + 3i, 1 - 3i]$; length 2 gives $[2, 2, 2, 6i]$; length 4 with twiddles $1, -i$ gives $[4, 8, 0, -4]$; divide by 4.

A13-F06

The result is $[3, 10, 8]$. At $N = 2$ the degree-2 coefficient wraps into index 0, yielding circular result $[11, 10]$.

A13-F07

The output has $7 + 10 - 1 = 16$ coefficients, so $N = 16$. The three transforms and pointwise product cost $O(N \log N)$ arithmetic operations.

A13-F08

If the true coefficient is known to be an integer and a proved absolute error bound is below 0.49 in this instance, it must be 17. The generic error-below-$1/2$ guarantee permits rounding each computed value to its nearest integer; an actual error bound or exact verification is needed.

A13-Z01 Solution

Over $\mathbb{F}_5$, $L_0 = 3(z - 1)(z - 2)$, $L_1 = 4z(z - 2)$ and $L_2 = 3z(z - 1)$ because $2^{-1} = 3$ and $(-1)^{-1} = 4$. Their weighted sum is $P = L_0 + 3L_1 + 2L_2 = 1 + z + z^2$; the three values are $1, 3, 2$. The determinant is $(1 - 0)(2 - 0)(2 - 1) = 2 \neq 0$, so the solution is unique.

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A13-Z02 Solution

$4^2 = 16 \neq 1$, $4^4 = 1$, and $4 \neq 1$, so its order is 4. The transform is $[2, 13, 0, 6]$. The inverse root and N^{-1} are both 13; the four unscaled inverse sums are $[4, 8, 0, 13]$, and multiplying by 13 gives $[1, 2, 0, 16]$. Since $16^2 = 1$, the element 16 has order 2 and its powers repeat before four points.

A13-Z03 Solution

There are four butterflies per stage and three stages, hence 12 butterflies, $\Theta(8 \log_2 8)$ work and three stages of depth, plus constant-depth initialization under the stated addressing assumption. A linear convolution of two length-3 sequences has length 5. A length-4 cyclic convolution wraps the degree-4 coefficient into the constant coefficient; the next usable length is 8.

A13-Z04 Solution

Evaluation at $1, i, -1, -i$ gives $\widehat{A} = [3, 1 + 2i, -1, 1 - 2i]$ and $\widehat{B} = [7, 3 + 4i, -1, 3 - 4i]$. Their pointwise product is $C = [21, -5 + 10i, 1, -5 - 10i]$. Using inverse root $-i$, the unscaled sums for $j = 0, 1, 2, 3$ are $[12, 40, 32, 0]$; scaling by $1/4$ gives $[3, 10, 8, 0]$.

A13-Z05 Solution

After bit reversal the original indices are $[0, 4, 2, 6, 1, 5, 3, 7]$. Stage 1 computes two-point DFTs of blocks $[0, 4]$, $[2, 6]$, $[1, 5]$, $[3, 7]$. Stage 2 computes four-point DFTs of $[0, 2, 4, 6]$ and $[1, 3, 5, 7]$. At stage 3 these are the even and odd subsequences, combined as $E_k \pm \omega_8^k O_k$. Sequential depth-first release uses $O(N)$ peak space; retaining complete arrays at all levels uses $\Theta(N \log N)$.

A13-Z06 Solution

All coefficients are modulo 5. $G = z(z-1)(z-2)$, $Q_0 = (z-1)(z-2)$, $Q_1 = z(z-2)$, $Q_2 = z(z-1)$; $d_0 = 2$, $d_1 = 4$, $d_2 = 2$, with inverses $3, 4, 3$. Thus $P = 1 \cdot 3Q_0 + 3 \cdot 4Q_1 + 2 \cdot 3Q_2 = 1 + z + z^2$. In general, building G costs $O(n^2)$, and each quotient, denominator evaluation and accumulation costs $O(n)$, for $O(n^2)$ field operations total. Both $P(z) = 0$ and $Q(z) = z(z-1)$ vanish at 0 and 1 and are distinct.

A13-Z07 Solution

Start with $b = 0$. At $k = 2$, $b = 4$; at $k = 1$, $b = 11$; at $k = 0$, $b = 24$. If the preceding value is $\sum_{j=k+1}^{2} a_j x^{j-k-1}$, the update gives $a_k + xb = \sum_{j=k}^{2} a_j x^{j-k}$. The empty-sum initialization and $k = 0$ termination prove correctness. For n coefficients this implementation takes $\Theta(n)$ field operations at one point and $\Theta(n^2)$ at n points separately.

A13-Z08 Solution

The first forward DFT is $[3, 1 + 2i, -1, 1 - 2i]$. A second forward transform with root i gives unscaled $[4, 0, 0, 8]$, and division by 4 gives $[1, 0, 0, 2]$. The root-of-unity sum proves the general result is $x_{(-j) \bmod 4}$; reversing those cyclic indices once more recovers the input. Alternatively apply the inverse root $-i$ and divide by 4 directly.

References and attribution

CLRS, Introduction to Algorithms, third edition, chapters 28 and 30. Matrix: PDF pages 834–861. FFT: PDF pages 919, 921, 923, 928–936, 939–941, 944–945 (PDF page numbering, third edition). MIT 6.046J Fall 2004 lecture 3 slides, PDF pages 27 and 35–43, and Fall 2005 lecture 3 notes support matrix multiplication and Strassen, not FFT.

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